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ECA5801-IOE AND AI DRIVEN SYSTEM

SIMULATION

EXPERIMENT 6.A

A.Simulation and Performance Evaluation of People–Process–Data–Things IoE Framework using Edge/Fog-to-Cloud Pipeline with MQTT, Node-RED.

CODE:

```
# ============================================================
# EXPERIMENT 6(a)
# Simulation and Performance Evaluation of P-P-D-T IoE
# Framework using Edge/Fog-to-Cloud Pipeline with MQTT,
# Node-RED and 5G/6G
# ============================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(42)

# ------------------------------------------------------------
# 1. P-P-D-T DATA FLOW
# ------------------------------------------------------------

ppdt = ["People", "Process", "Data", "Things"]

print("========== P-P-D-T IoE DATA FLOW ==========\n")
```

```
print(" → ".join(ppdt))
print("\nPeople → Process → Data → Things")
print("            ↓")
print("          MQTT")
print("            ↓")
print("    Edge → Fog → Cloud")
print("            ↓")
print("        Node-RED")

# ------------------------------------------------------------
# 2. EDGE / FOG / CLOUD PARAMETERS
# ------------------------------------------------------------

layers = ["Edge", "Fog", "Cloud"]

# Processing capacity in Mbps
capacity = {
   "Edge": 100,
   "Fog": 75,
   "Cloud": 50
}

# Base latency in milliseconds
base_latency = {
   "Edge": 10,
   "Fog": 25,
   "Cloud": 60
}
```

```
# Packet loss probability
loss_rate = {
    "Edge": 0.02,
    "Fog": 0.04,
    "Cloud": 0.06
}

# ------------------------------------------------------------
# 3. SIMULATION
# ------------------------------------------------------------

total_packets = 1000
packet_size_kb = 2
simulation_time = 10

results = []

for layer in layers:

    lost_packets = int(total_packets * loss_rate[layer])
    delivered_packets = total_packets - lost_packets

    # Data successfully delivered
    data_bits = delivered_packets * packet_size_kb * 8 * 1024

    # Throughput
    throughput = data_bits / simulation_time / 1e6

    # Limit according to layer capacity
```

```
    throughput = min(throughput, capacity[layer])

    # Latency
    latency = base_latency[layer] + np.random.uniform(2, 8)

    results.append([
        layer,
        total_packets,
        delivered_packets,
        throughput,
        latency
    ])

# ------------------------------------------------------------
# 4. RESULT TABLE
# ------------------------------------------------------------

df = pd.DataFrame(
    results,
    columns=[
        "Layer",
        "Total Packets",
        "Delivered Packets",
        "Throughput (Mbps)",
        "Latency (ms)"
    ]
)

df["Throughput (Mbps)"] = df["Throughput (Mbps)"].round(2)
```

```
df["Latency (ms)"] = df["Latency (ms)"].round(2)

print("\n========== EDGE / FOG / CLOUD RESULTS ==========\n")
display(df)

# ------------------------------------------------------------
# 5. THROUGHPUT GRAPH
# ------------------------------------------------------------

plt.figure(figsize=(6, 4))

plt.bar(
    df["Layer"],
    df["Throughput (Mbps)"]
)

plt.title("Edge / Fog / Cloud - Throughput")
plt.xlabel("Processing Layer")
plt.ylabel("Throughput (Mbps)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------------------------------------
# 6. LATENCY GRAPH
# ------------------------------------------------------------

plt.figure(figsize=(6, 4))
```

```
plt.bar(
    df["Layer"],
    df["Latency (ms)"]
)

plt.title("Edge / Fog / Cloud - Latency")
plt.xlabel("Processing Layer")
plt.ylabel("Latency (ms)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------------------------------------
# 7. 5G vs 6G COMPARISON
# ------------------------------------------------------------

network = ["5G", "6G"]

network_throughput = [1000, 5000]
network_latency = [10, 1]

network_df = pd.DataFrame({
    "Network": network,
    "Throughput (Mbps)": network_throughput,
    "Latency (ms)": network_latency
})

print("\n========== 5G vs 6G ==========\n")
display(network_df)
```

```
# ------------------------------------------------------------
# 8. 5G / 6G GRAPHS
# ------------------------------------------------------------

fig, ax = plt.subplots(1, 2, figsize=(9, 4))

ax[0].bar(
    network_df["Network"],
    network_df["Throughput (Mbps)"]
)

ax[0].set_title("5G vs 6G Throughput")
ax[0].set_ylabel("Mbps")

ax[1].bar(
    network_df["Network"],
    network_df["Latency (ms)"]
)

ax[1].set_title("5G vs 6G Latency")
ax[1].set_ylabel("ms")

plt.tight_layout()
plt.show()

# ------------------------------------------------------------
# 9. CONCLUSION
# ------------------------------------------------------------
```

```
print("\n========== CONCLUSION ==========\n")

print("1. Edge processing provides lower latency because")
print("   processing occurs close to IoE devices.")

print("\n2. Fog acts as an intermediate layer between Edge and Cloud.")

print("\n3. Cloud processing generally introduces higher latency")
print("   because data travels farther.")

print("\n4. In this simulation, 6G provides higher throughput")
print("   and lower latency than 5G.")

print("\n5. MQTT is used as the communication concept and")
print("   Node-RED represents the IoE data-flow/workflow layer.")
```

OUTPUT:

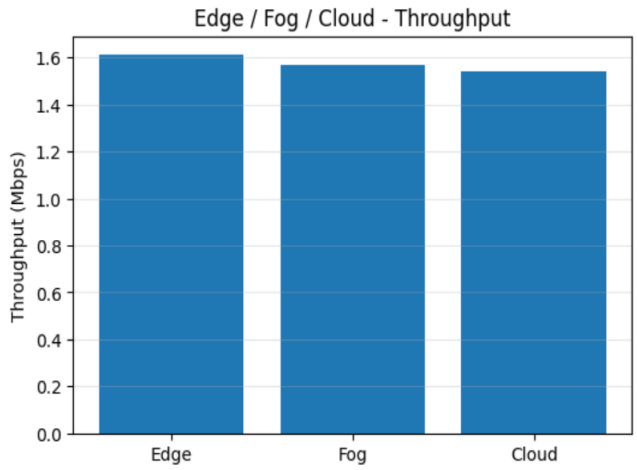
```
========== P-P-D-T IoE DATA FLOW ==========

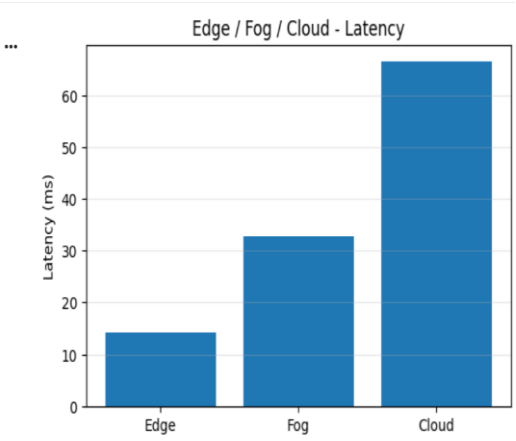
People → Process → Data → Things

People → Process → Data → Things
                    ↓
                  MQTT
                    ↓
         Edge → Fog → Cloud
                    ↓
               Node-RED

========== EDGE / FOG / CLOUD RESULTS ==========
```

	Layer	Total Packets	Delivered Packets	Throughput (Mbps)	Latency (ms)
0	Edge	1000	980	1.61	14.25
1	Fog	1000	960	1.57	32.70
2	Cloud	1000	940	1.54	66.39






```
========== 5G vs 6G ==========
```

	Network	Throughput (Mbps)	Latency (ms)
0	5G	1000	10
1	6G	5000	1

```
========== CONCLUSION ==========

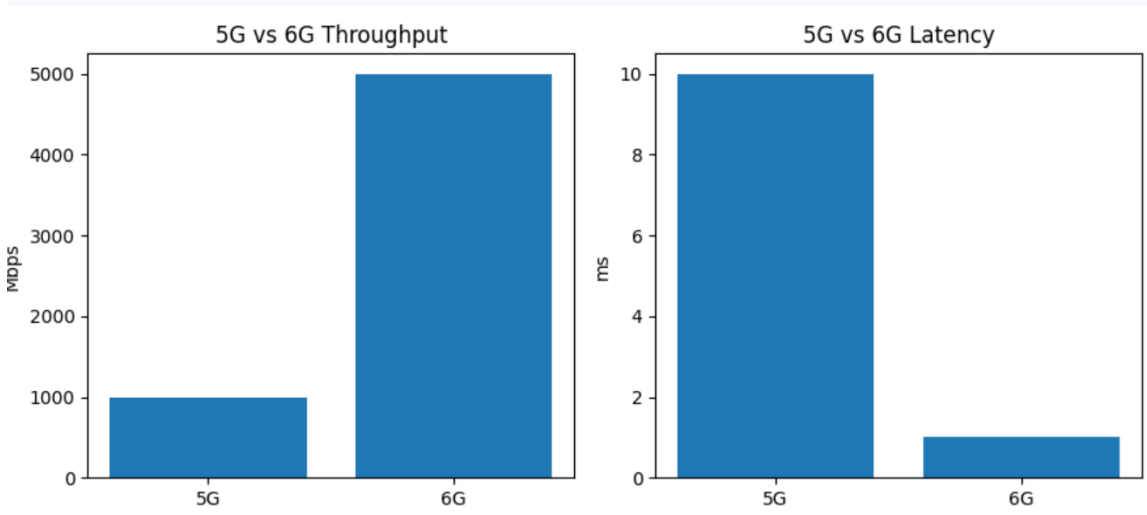
1. Edge processing provides lower latency because
   processing occurs close to IoE devices.

2. Fog acts as an intermediate layer between Edge and Cloud.

3. Cloud processing generally introduces higher latency
   because data travels farther.

4. In this simulation, 6G provides higher throughput
   and lower latency than 5G.

5. MQTT is used as the communication concept and
```


5G vs 6G Throughput
Mbps
0
1000
2000
3000
4000
5000
5G
6G
5G vs 6G Latency
ms
0
2
4
6
8
10
5G
6G

EXPERIMENT 6.B

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B.Benchmarking of 5G/6G Communication for IoE Applications based on Throughput, Latency, Reliability and Energy Efficiency

CODE:

```
# ============================================================
# EXPERIMENT 6(b)
# Benchmarking of 5G/6G Communication for IoE Applications
# based on Throughput, Latency, Reliability and Energy Efficiency
# ============================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ------------------------------------------------------------
# 1. NETWORKS
# ------------------------------------------------------------

networks = ["5G", "6G"]

# Representative simulation parameters
throughput = [1000, 5000]        # Mbps
latency = [10, 1]                # ms
reliability = [99.0, 99.99]      # %
energy_efficiency = [70, 90]     # %

# ------------------------------------------------------------
# 2. RESULT TABLE
# ------------------------------------------------------------
```

```
df = pd.DataFrame({
    "Network": networks,
    "Throughput (Mbps)": throughput,
    "Latency (ms)": latency,
    "Reliability (%)": reliability,
    "Energy Efficiency (%)": energy_efficiency
})

print("========== 5G vs 6G BENCHMARK RESULTS ==========\n")

display(df)

# ------------------------------------------------------------
# 3. THROUGHPUT
# ------------------------------------------------------------

plt.figure(figsize=(6, 4))

plt.bar(
    df["Network"],
    df["Throughput (Mbps)"]
)

plt.title("5G vs 6G - Throughput")
plt.xlabel("Network")
plt.ylabel("Throughput (Mbps)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------------------------------------
```

```
# 4. LATENCY
# ------------------------------------------------------------

plt.figure(figsize=(6, 4))

plt.bar(
    df["Network"],
    df["Latency (ms)"]
)

plt.title("5G vs 6G - Latency")
plt.xlabel("Network")
plt.ylabel("Latency (ms)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------------------------------------
# 5. RELIABILITY
# ------------------------------------------------------------

plt.figure(figsize=(6, 4))

plt.bar(
    df["Network"],
    df["Reliability (%)"]
)

plt.title("5G vs 6G - Reliability")
plt.xlabel("Network")
plt.ylabel("Reliability (%)")
```

```
plt.ylim(95, 100)

plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------------------------------------
# 6. ENERGY EFFICIENCY
# ------------------------------------------------------------

plt.figure(figsize=(6, 4))

plt.bar(
    df["Network"],
    df["Energy Efficiency (%)"]
)

plt.title("5G vs 6G - Energy Efficiency")
plt.xlabel("Network")
plt.ylabel("Energy Efficiency (%)")

plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------------------------------------
# 7. COMBINED GRAPH
# ------------------------------------------------------------

fig, ax = plt.subplots(2, 2, figsize=(9, 7))
```

```
ax[0, 0].bar(
    df["Network"],
    df["Throughput (Mbps)"]
)
ax[0, 0].set_title("Throughput")
ax[0, 0].set_ylabel("Mbps")

ax[0, 1].bar(
    df["Network"],
    df["Latency (ms)"]
)
ax[0, 1].set_title("Latency")
ax[0, 1].set_ylabel("ms")

ax[1, 0].bar(
    df["Network"],
    df["Reliability (%)"]
)
ax[1, 0].set_title("Reliability")
ax[1, 0].set_ylabel("%")
ax[1, 0].set_ylim(95, 100)

ax[1, 1].bar(
    df["Network"],
    df["Energy Efficiency (%)"]
)
ax[1, 1].set_title("Energy Efficiency")
ax[1, 1].set_ylabel("%")

plt.tight_layout()
plt.show()
```

```
# ------------------------------------------------------------
# 8. AUTOMATIC COMPARISON
# ------------------------------------------------------------

print("\n========== PERFORMANCE COMPARISON ==========\n")

if throughput[1] > throughput[0]:
    print("✓ 6G has higher throughput than 5G.")

if latency[1] < latency[0]:
    print("✓ 6G has lower latency than 5G.")

if reliability[1] > reliability[0]:
    print("✓ 6G has higher reliability than 5G.")

if energy_efficiency[1] > energy_efficiency[0]:
    print("✓ 6G has better energy efficiency than 5G.")

# ------------------------------------------------------------
# 9. FINAL CONCLUSION
# ------------------------------------------------------------

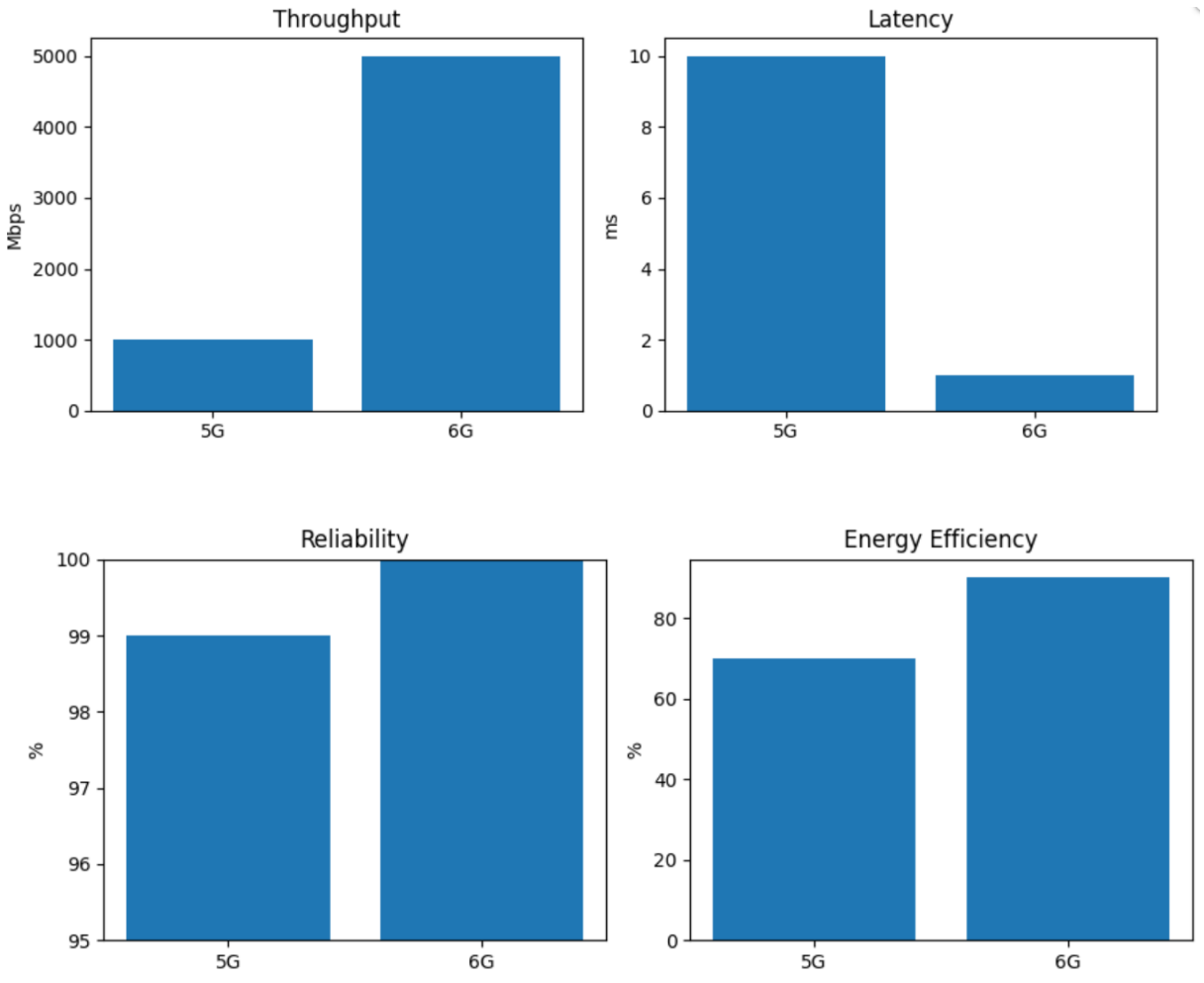
print("\n========== CONCLUSION ==========\n")

print("6G shows better simulated performance than 5G")
print("for IoE applications in this benchmark.")
print("\n• Higher throughput")
print("• Lower latency")
print("• Higher reliability")
print("• Better energy efficiency")
```

OUTPUT:

```
========== 5G vs 6G BENCHMARK RESULTS ==========
```

	Network	Throughput (Mbps)	Latency (ms)	Reliability (%)	Energy Efficiency (%)
0	5G	1000	10	99.00	70
1	6G	5000	1	99.99	90



========== PERFORMANCE COMPARISON ==========

✓ 6G has higher throughput than 5G.
✓ 6G has lower latency than 5G.
✓ 6G has higher reliability than 5G.
✓ 6G has better energy efficiency than 5G.

========== CONCLUSION ==========

6G shows better simulated performance than 5G
for IoE applications in this benchmark.

- Higher throughput
- Lower latency
- Higher reliability
- Better energy efficiency